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To cite this article: Hill Kulu, Bernice Kuang, Sarah Christison & Ann Berrington (2026) Long-term fertility trends by birth order in Britain: Comparison between England & Wales and Scotland, *Population Studies*, 80:1, 1-23, DOI: [10.1080/00324728.2025.2491354](https://doi.org/10.1080/00324728.2025.2491354)

To link to this article: <https://doi.org/10.1080/00324728.2025.2491354>



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Published online: 12 Jun 2025.



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Long-term fertility trends by birth order in Britain: Comparison between England & Wales and Scotland

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This study uses census-linked administrative data to investigate childbearing trends by birth order in Britain over three decades. This is the first study to investigate longer-term changes in fertility dynamics in Britain by birth order and to compare parity-specific fertility by country. First-birth rates declined in the 1990s, slightly increased in the first decade of this century, and decreased thereafter, with changes in timing of parenthood responsible for these changes. Second- and third-birth rates declined in the 1990s but remained relatively stable in the early twenty-first century. Birth intervals remained unchanged, meaning that changes in quantum are responsible for trends in higher-order birth rates. Time trends are similar in England & Wales and Scotland but with significantly lower second- and third-birth rates in Scotland. Changes in population composition by education and ethnicity explain little of the aggregate fertility trends or between-country differences. Both countries have seen rapid declines in first-birth rates, especially among low-educated women.

Keywords: fertility; parity; education; ethnicity; Britain

[Submitted March 2024; Final version accepted December 2024]

Context

The first two decades of the twenty-first century witnessed significant changes in fertility in Europe. While period fertility increased in the first decade (the 2000s), especially in Northern and Western Europe, fertility declined significantly in the second decade (the 2010s) and reached historically low levels in many countries (Hellstrand et al. 2020; Ermisch 2021, 2024; Berrington et al. 2022; Ohlsson-Wijk and Andersson 2022). Most studies investigating recent fertility decline in Europe have emphasized the importance of economic uncertainty (Bueno 2020; Alderotti et al. 2021; Matysiak et al. 2021; Savelieva et al. 2023), but there has also been research underlining the role of ideational change (Lesthaeghe and Lopez-Gay 2013; Lesthaeghe 2014), gender inequalities (Goldscheider et al. 2015), and welfare state policies and provisions (Neyer and Andersson 2008; Thévenon and Gauthier 2011; Gauthier 2013). This paper provides new insights into long-term changes

in fertility dynamics in Britain, presenting the first detailed comparisons between Scotland and England & Wales and for population subgroups by education and ethnicity. We argue that to distinguish between wider explanations for fertility change, we must first understand more precisely the changing demographic components of childbearing from a parity perspective.

Measures of aggregate period fertility reflect both the timing and level of childbearing (tempo and quantum, respectively). For example, the Total Fertility Rate (TFR) may decline due to postponement of family formation or due to increased childlessness. A decline may also be caused by decreasing higher-order birth rates, because of either a lengthening of birth intervals or a decline in completed family size. Similarly, the TFR may increase due to a stopping of postponement, a decline in childlessness, a shortening of birth intervals (as in Sweden in the early 1990s; see Hoem 1990), or an increase in completed family size. Fertility timing and levels are often closely related, although not always: for

example, the postponement of family formation is likely to lead to a decline in family size, although not necessarily: small family sizes are nonetheless possible in a context of early childbearing and low childlessness. Therefore, identifying exact changes in childbearing behaviour is of critical importance for understanding fertility dynamics. Period fertility may also decline or increase because of changes in population composition. The role of increased educational enrolment rates and educational qualifications (driven by wider societal changes) in declining period fertility is a well-known example (Ní Bhrolcháin and Beaujouan 2012). By contrast, an increased number of immigrants and ethnic minorities in a population is often considered a factor behind increased period fertility (Sobotka 2008; Tromans et al. 2009).

This study investigates childbearing trends in Britain—England & Wales and Scotland—by birth order over the last three decades. The novel contributions of the study are as follows. First, we determine whether changes in the TFR are attributable to changes in first, second, third, or higher-order births, or all of these. Second, we examine whether and how changes in fertility timing explain fertility trends by parity. Third, we compare fertility trends by parity in England & Wales vs Scotland. Finally, we conduct separate analyses by educational level and ethnic origin to understand whether and how changes in population composition explain fertility trends over time and differences across countries.

Most analyses of fertility trends use aggregate fertility measures, which are easy to calculate, understand, and relate to wider economic, social, political, and cultural trends. However, identifying and understanding behavioural changes requires analysis of fertility trends and patterns by birth order (Ní Bhrolcháin 1987; Rendall and Smallwood

2003). Childbearing is essentially a sequential process. Individuals may hold preferences for a specific family size, but the decision to have a child is made separately for each child. This will depend on both individuals' and couples' preferences and socio-economic circumstances. Although recent fertility change in Britain has been studied in detail, the focus has been on England and on the last decade (Ermisch 2021, 2024). Britain offers a unique opportunity to compare fertility trends and patterns in countries with similar institutional, political, and economic environments but distinct national identities. Such a comparison is novel in the recent fertility literature, in which studies have compared fertility either across European countries (Brini 2020) or across geographical units within countries (Matysiak et al. 2021; Ohlsson-Wijk and Andersson 2022).

Explaining period fertility in Britain

In 2001, the TFR in England & Wales was 1.63. In 2012, after a decade of increase, the figure was 1.93. This consistent increase was particularly noticeable, given that in most other European countries, fertility had fallen during the economic downturn of 2008–11 (Sobotka et al. 2011). Since 2012, total fertility has declined, to 1.44 in 2023 (ONS 2024). Aggregate fertility dynamics have been similar in the three constituent countries of Britain, but the levels have varied significantly. Total fertility in Scotland has been around 10 per cent lower than in England or Wales (which are more similar to each other) (Figure 1). In 2023, the TFR was 1.39 in England, 1.30 in Wales, and 1.30 in Scotland (NRS 2024; ONS 2024).

There are several competing demographic explanations for the changes in fertility in Britain since the

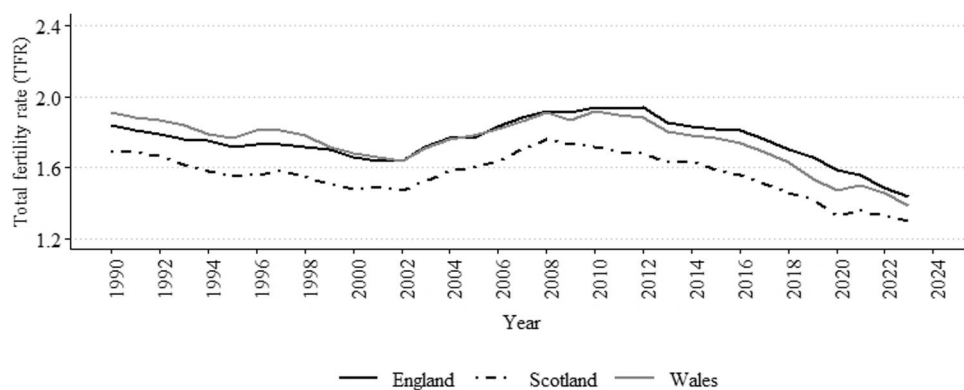


Figure 1 Total fertility in Britain, 1990–2023

Source: NRS 2024; ONS 2024.

1990s: changes in the timing of family formation, changes in the quantum of higher-order births and the intervals between them, and changes in population composition.

Changes in the timing of family formation

The TFR shows the average number of children for a hypothetical cohort of women in a given period of time. However, if mean age at childbearing increases, the TFR underestimates cohort fertility (Bongaarts and Feeney 1998). A stopping of fertility postponement (or a deceleration of the rate of postponement) leads to an increase in the TFR. Postponement of childbearing can thus be seen as the cause of low fertility in Europe in the 1990s, and the end of fertility postponement can be viewed as the reason for the upturn experienced by many European countries, including Britain, in the 2000s (Bongaarts and Sobotka 2012). A new or further postponement could explain the significant fertility decline since 2012. The role of timing in explaining period fertility trends has been indirectly supported by recent studies on Britain and Nordic countries showing that the decline in first-birth rates explained most of the decline in TFR during the 2010s (Hellstrand et al. 2020; Ermisch 2021; Ohlsson-Wijk and Andersson 2022). Postponement of childbearing, rather than increased childlessness, is very likely the main reason for the decline in first-birth rates. However, analyses of fertility by cohort have shown that in Finland, women born in the late 1970s and 1980s are projected to experience lower lifetime fertility than earlier cohorts (Hellstrand et al. 2020). This suggests that changes in the timing of family formation may account for a substantial part of, but not all, recent changes in fertility rates in Britain.

Changes in higher-order births: Quantum and intervals

The upturn in the TFR in the 2000s may have been due to behavioural changes related to second and higher-order births, with women more likely to have another child or to have another child after a shorter birth interval. It is possible that childbearing behaviour responded to family-friendly policies introduced in the late 1990s (including improved parental leave policies and an expansion of childcare) that reduced the opportunity costs of childrearing, especially for more highly educated women (Sigle-

Rushton 2008). While increased availability of good-quality childcare should reduce the work–family conflict for all women, the reduction of ‘opportunity costs’ and ‘wage penalties’ should be felt most by highly educated women with stronger ties to the labour market, thus precipitating a narrowing of educational differences in childbearing. We might also expect to observe some changes in the spacing of birth intervals, although the direction of such changes is more difficult to predict. On the one hand, increased availability of high-quality (subsidized) childcare is expected to reduce the pressure to have children within a short time interval, as was the case in the UK until recently, when working women had to rely on a combination of expensive childcare offered by the market and informal childcare provided by grandparents or friends (with children) (Ekert-Jaffé et al. 2002). On the other hand, with increased labour force participation rates among women, a growing share of women may not wish to prolong the period of raising young children, which, despite increased availability of childcare, is still full of interruptions and complex logistics (Sigle-Rushton 2008). This would be supported by decreasing differences in birth intervals between educational groups and a shortening of birth intervals for all women.

Changes in population composition

Increased enrolment in higher education in the 1990s resulted in postponement of childbearing in European countries and declining period fertility (Frejka and Sobotka 2008; Neels and de Wachter 2010; Ní Bhrolcháin and Beaujouan 2012). UK enrolment rates increased only slowly during the 2000s, which may explain the cessation of postponement (before the further rapid declines in teenage fertility since 2012). With the increased enrolment rates in the 1990s, the average educational level of British women increased. As highly educated UK women exhibit lower completed fertility than women with medium or low educational qualifications (Kneale and Joshi 2008; Berrington et al. 2015), the upturn in total fertility in the first decade of this century was surprising in light of the educational changes. It is possible that other compositional changes (e.g. an increased number of immigrants) outweighed the fertility-lowering effect of educational changes. Alternatively, the fertility rates of highly educated women may have increased in the 2000s because of a deceleration of postponement and declining fertility differences by

educational qualifications, although research to date has provided no strong support to the idea of fertility recuperation among highly educated British women at older ages (Berrington et al. 2015). In contrast, the fertility decline since 2012, possibly due to postponement of childbearing, would be very consistent with the changes in educational composition of the British population (increased enrolment rates and average educational levels for women).

Immigration to the UK and other European countries has increased significantly since the turn of the century (Raymer et al. 2011). Migrants contribute to increased fertility in the following ways. First, many come from countries where family size is declining but still high; although the fertility behaviour of migrants changes when they move from one country to another, those coming from high-fertility countries still have larger families on average than women from the destination country (Andersson and Scott 2007; Kulu et al. 2017). Second, many female migrants arrive because of marriage or family reunification and have a (first) child soon after migration (and marriage), supporting the well-known relationship between marriage and family formation (Andersson 2004; Kulu 2005; Robards and Berrington 2016). The TFR may thus be high not only for immigrant women from high-fertility countries but also for marriage migrants from low-fertility countries, because partnered women with short marriage durations are over-represented among them (or in the denominator of the formula for calculating the TFR) (Toulemon 2004). Increased numbers of immigrants in the UK may therefore explain the simultaneous upturn of total fertility in the 2000s (Tromans et al. 2009).

An increasing number of descendants of post-war labour migrants may also explain the upturn in fertility experienced by Western and Northern Europe, including Britain, in the first decade of this century. Although fertility rates among descendants of immigrants are usually lower than those of immigrants, for some ethnic groups (e.g. Bangladeshi and Pakistani in Britain) fertility is still considerably higher among descendants of immigrants than among women without immigrant parents (or grandparents) (Coleman and Dubuc 2010; Kulu and Hannemann 2016). The increased numbers of immigrants and their descendants suggest that period fertility should have continued to stay relatively high during the 2010s. However, as fertility has declined since 2012, the reasons for this decline may lie elsewhere.

Recent changes

The post-2008 economic recession and subsequent austerity are likely to account for some of the recent fertility decline in Europe. Rising unemployment, economic uncertainty, and lack of housing affordability may have directly shaped couples' childbearing plans and behaviour. Further, large cuts in public spending in European countries (including the UK) in the 2010s reduced social and family-related expenditure and signalled to families that times were hard, creating a pessimistic climate (Sobotka et al. 2011). We expect to observe not only postponement of childbearing but potentially also increasing birth intervals since 2008, or at least after 2012 (i.e. as couples waited to see how the economic climate would evolve). On the one hand, we might expect fertility rates to have declined most among highly educated women, who may perceive childbearing to be an especially risky strategy during recession and economic uncertainty due to very high opportunity costs (Sobotka et al. 2011). On the other hand, recession and subsequent austerity may have hit the lowest educational group hardest; fertility declines at the youngest ages in the UK since 2010 provide some support for this argument (Ermisch 2021; Berrington et al. 2022). Recent research from Nordic countries also supports the theory that fertility decline has accelerated among the least educated: those with weaker labour market attachment and lower earnings (Ohlsson-Wijk and Andersson 2022). Further, fertility has declined significantly not only among those without a degree but also among those with a degree in a field characterized by higher economic uncertainty (Hellstrand et al. 2022).

Country differences

At aggregate level, fertility rates tend to be lower in Scotland than in England & Wales (with fertility in Wales relatively similar to that in England). Fertility was higher in Scotland than England in the 1960s and 1970s; however, the fertility decline in Scotland was steeper, and since the early 1980s, total fertility has been consistently lower in Scotland than in England & Wales. For example, in 2023, the TFR was 1.30 in Scotland, 1.39 in Wales, and 1.44 in England. The reasons for the country differences are not clear. Previous research found that first-birth rates were relatively similar in England and Scotland but that greater delays to second and

third births were characteristic of Scotland's relatively low fertility (Graham et al. 2007). That study emphasized the longer birth intervals in Scotland, although it remains unclear whether longer birth intervals ultimately led to lower second- and third-birth levels. Their analysis of fertility intentions showed no evidence that Scottish women intended to have fewer children than English women. However, a recent study by Berrington et al. (2023) showed that while there were no differences in intentions to have a first birth, Scottish mothers were slightly less likely to hold a firm intention to have additional births than mothers in England or Wales. It is also possible that fertility is lower in Scotland than in England because relatively fewer immigrants and their descendants live in Scotland.

Expectations

To sum up, this study investigates the impacts of the timing of family formation, the timing and quantum of higher-order births, and population composition on childbearing dynamics in Britain over the past three decades. Based on previous research, we first expect to observe significant fluctuation in first-birth rates, likely driven by changes in the timing of family formation. Second, we expect to observe a change in second-, third-, and also fourth-birth rates, which would largely follow patterns similar to first births: a decline in the 1990s and increase in the 2000s, followed by a new decline in the 2010s. However, it is less clear how much the higher-order birth rates would vary over time. It is also less clear whether these trends occur mostly because of changes in birth intervals or changes in higher-order birth levels. Third, we expect to observe similar first-birth rates in England & Wales and in Scotland but lower second-, third-, and fourth-birth rates in Scotland. Again, it is important to determine whether Scottish women simply experience longer birth intervals or whether they are also less likely to ever have a second, third, and fourth child. Finally, we expect changes in population composition to play a limited role in recent fertility fluctuations; most educational and ethnic groups are expected to follow largely similar trends, although it is likely that the recent decline in fertility was accelerated among disadvantaged population groups, as in Nordic countries. We are convinced that a detailed examination of the childbearing trends across population subgroups will improve our understanding of how compositional and other

factors have shaped the fertility behaviour of women in Britain.

Methods

We examine childbearing trends by birth order; this is necessary to detect underlying changes in fertility patterns (Ní Bhrolcháin 1987; Murphy and Berrington 1993; Smallwood 2002). We calculate annual parity-specific fertility rates using hazard regression (or survival analysis) (Hoem 1987; Kulu and Hanne-mann 2016; Ohlsson-Wijk and Andersson 2022). Our analytical strategy is as follows. For first births, we calculate annual first-birth rates standardized for exact age of woman by fitting a proportional hazards regression model with three covariates: women's age, calendar time, and country (England & Wales, Scotland). For second, third, and fourth births, we calculate annual parity-specific rates standardized also for duration since last birth (Hoem 1990; Andersson 1999; Kulu and Hannemann 2016). This allows us to determine whether the recent changes in the TFR are attributable to changes in first-, second-, third-, or fourth-birth rates, or all four, and whether this is similar for England & Wales and Scotland.

Next, we calculate annual first-birth rates by women's age. This provides us with valuable information on whether the (expected) increase in first-birth rates in the 2000s is explained by the stopping of postponement combined with the 'recuperation' effect or rather by an increased share of women having children. Similarly, we calculate duration-specific rates for second and third births to detect a possible acceleration of childbearing in the 2000s and deceleration in the 2010s. We examine whether there has been a shortening/lengthening of birth intervals or an increase/decrease in completed family size.

The basic model can be formalized as follows:

$$\ln h_i(t) = \ln h_0(t) + \sum_l \alpha_l x_{il}(t) + \beta z_i, \quad (1)$$

where $h_i(t)$ denotes the hazard of having an n th birth for individual i at duration t ; $h_0(t)$ is the baseline risk, by either women's age (for first births) or time since previous birth (for second, third, or fourth births); $x_{il}(t)$ denotes time-varying and time-constant covariates, such as calendar year and age at previous birth (if any), with parameters α_l measuring their effect; and z_i is a variable for country, with β measuring its impact.

However, the individual-level administrative data used in this paper must be analysed within a trusted research environment and cannot be merged due to data confidentiality. To overcome this issue, we use a count data approach to explicitly compare parity-specific fertility rates across Britain's countries adjusted by various covariates (Hoem 1987; Preston 2005).

We can specify the baseline hazard as a piecewise constant:

$$\ln h_0(t) = \ln h_j \quad \text{for } t \text{ in } [t_j, t_{j+1}). \quad (2)$$

The basic model in equation (1) then becomes as follows:

$$\ln h_{jm} = \ln h_j + \sum_l \alpha_l x_{jml} + \beta z_{jm}, \quad (3)$$

where h_{jm} is the hazard for time interval j (e.g. age group) and (covariate) category m . Holford (1980) and Laird and Olivier (1981) have demonstrated that log-linear models for the cell means of contingency tables with Poisson data are equivalent to log-linear hazard models for survival data with piecewise constant baseline hazard and categorical covariates. We use the individual-level data to calculate aggregated event-time (occurrence-exposure) tables for each country, which are aggregated by different combinations of variables; the country files are then merged into one common database and the data analysed using a Poisson regression model (Kulu et al. 2017). Using this approach, we can explicitly compare childbearing trends by parity in England & Wales vs Scotland.

Data and their quality

ONS Longitudinal Study and Scottish Longitudinal Study

Vital statistics are useful for calculating aggregate fertility measures but do not provide sufficient information to examine fertility dynamics by parity; until 2012 in England & Wales and 2013 in Scotland, information on birth order was collected only for births occurring within marriage. Although information on birth order is now available for all births, there are no annual data on the risk population by parity. The risk population must be estimated using survey data (e.g. the Annual Population Survey). Also, only limited information is available from vital statistics on individuals' socio-economic characteristics. Thus, many

researchers have analysed fertility by parity using longitudinal household survey data instead (Ermisch 2021, 2024). However, the analysis of small-scale survey data, sometimes for a specific cohort, has its own limitations. Most importantly, such data rarely provide researchers with reliable information on childbearing trends over time: there is considerable imprecision in estimated fertility trends by birth order, and analysis of trends in parity-specific fertility by population subgroups (e.g. education) is not feasible. Further, existing longitudinal household surveys do not have large enough sample sizes for cross-national comparison.

We use data from the Office for National Statistics (ONS) Longitudinal Study (LS), and the Scottish Longitudinal Study (SLS). The LS was initiated in 1971 and contains linked census and vital events data on a 1 per cent sample of the population of England & Wales. This data set of currently more than 1 million records is attractive for fertility research as it contains full fertility histories for women as far back as those aged 15 in 1971, continuing with those becoming 15 in subsequent years. The LS thus allows us to calculate parity-specific birth rates, standardized for demographic characteristics (women's age and time since last birth, if any) and for characteristics collected in the census (education, ethnicity) (Robards et al. 2011). The SLS was initiated in 1991 and contains linked census and vital events data on a 5.3 per cent sample of the population of Scotland. It currently holds data on more than 250,000 individuals. The data sets allow the construction of fertility histories for women who were interviewed in 1991 and those who became 16 in subsequent years. Like the LS, the SLS thus allows us to calculate parity-specific birth rates over the past three decades standardized for demographic characteristics and characteristics collected in the Scottish census.

Study design and sample

Our study sample consists of women born in 1956 and later, who are followed from age 15 in England & Wales and 16 in Scotland. In both LSs, births have currently been linked up to 2017. From the LS we can construct fertility histories for all women born in 1956 and later. From 2001, when the oldest women in our sample reached age 45, we know the parity of (most) women of reproductive age in England & Wales. We can also calculate

Table 1 Number of women and births in the study sample

	Women (risk population)	Births
<i>First births</i>		
England & Wales	167,050	92,060
Scotland	53,301	23,352
<i>Second births</i>		
England & Wales	90,930	64,840
Scotland	29,873	18,039
<i>Third births</i>		
England & Wales	60,750	19,880
Scotland	24,372	6,585
<i>Fourth births</i>		
England & Wales	18,959	6,200
Scotland	6,772	1,447

Source: Authors' calculations using ONS LS and SLS.

parity-specific fertility rates earlier than this (e.g. from the 1990s, when the oldest women in the sample were in their late 30s), thus covering most women of reproductive age. The SLS began two decades later, in 1991. Fortunately, we can determine the parity of all Scottish women aged 16–49 from as early as 1991. In SLS data, household members can be linked and their relationship (e.g. a mother and her children) can be identified at each census year (including 1991). Still, some caution is needed with the application of the own-children method, because for older women some children might already have left the parental home. Therefore, we calculate parity only for women aged 16–35 in 1991, assuming that the children of most women of that age would still live with them. Nevertheless, parity-specific fertility rates can also be calculated for Scotland from the 1990s.

Individuals enter the study when they reach age 15 (or 16 in Scotland) or arrive in the UK. We exclude immigrants who arrive aged 15 (or 16 in Scotland) or older from the study as we cannot determine their parity. We thus analyse the fertility only of UK-born women and of immigrants who arrive aged 0–15. Individuals exit the study at age 50 or through death, emigration, or loss to follow-up. Individuals who are present at one census but absent at the subsequent census are censored at the midpoint between censuses. We exclude a small number of women who are not present at any census (but for whom there is information on events, such as births) or are untraced (present at census but no information on events). We include individuals' education and ethnicity in our analysis. We use the following educational categories: High (degree or higher), Medium (A-Levels or Highers), and Low

(GCSEs, National Fives, or lower). Ethnicity is defined as White British or Other. The latter group includes individuals of all other ethnicities, whether they are UK born or immigrants who arrived in the UK during childhood.

For first births, the risk population is 167,050 women in England & Wales and 53,301 in Scotland; for second births, the risk populations are 90,930 and 29,873; for third births 60,750 and 24,372; and for fourth births 18,959 and 6,772, respectively (Table 1). There are sufficient events to conduct detailed analysis of fertility trends across years by parity.

Data quality

The quality of the ONS LS and SLS data can be assessed. We can compare estimates from vital statistics with those from the two LSs (e.g. on aggregate fertility trends or mean family size by birth cohort). Our approach is to exploit both vital statistics and census data because of the uncertainty around the size of Britain's risk population when calculating demographic measures. Estimates based on census data should be more accurate: in census years, the discrepancy between the risk populations from different sources should be minimal, as census data are used to correct estimates of the denominators (i.e. population estimates) for vital rates. Nevertheless, we calculate age-specific fertility rates and the TFR for England & Wales and for Scotland in 2011 using the 2011 Census data (10 per cent microdata sample), vital statistics, and the ONS LS and SLS. Our analysis (not shown) found that overall, the TFR estimates from alternative sources were relatively similar, but the two LSs may slightly

underestimate fertility rates. The reason for the discrepancy might be that some births to the youngest women have not been linked in the LSs, and some people who give birth in a census year are not captured by the census (for further details, see Ellison et al. 2023).

Fertility trends in Britain by birth order and country

First births

Figure 2 presents relative first-birth rates over the last three decades, separately for England & Wales and Scotland. The rates are standardized by women's age and presented relative to a reference point: fertility in England & Wales in 2000.

We can see that first-birth rates declined in the 1990s, then increased in the 2000s, followed by a new decline in the latest decade. The trends are very similar in England & Wales and in Scotland.

Next, we disaggregate the first-birth rates by age. For England & Wales, first-birth rates at all ages declined in the 1990s (Figure 3); this was part of a longer-term decline. In the 2000s, the decline continued at ages 15–19, whereas first-birth rates at most other ages increased. We observe diverging trends from the beginning of the 2010s: first-birth rates declined at all ages below 30 or even 35 but increased at ages 35+. Clearly, it seems that the postponement of childbearing, the subsequent stopping of postponement (or a recuperation), and then a further postponement have characterized first-birth trends in England & Wales over the past three decades.



Figure 2 Relative first-birth rates in Britain, 1990–2017 (adjusted for women's age)

Notes: Reference group: England & Wales, 2000. Vertical bars show 95 per cent confidence intervals (CIs).

Source: Authors' calculations using ONS LS and SLS.

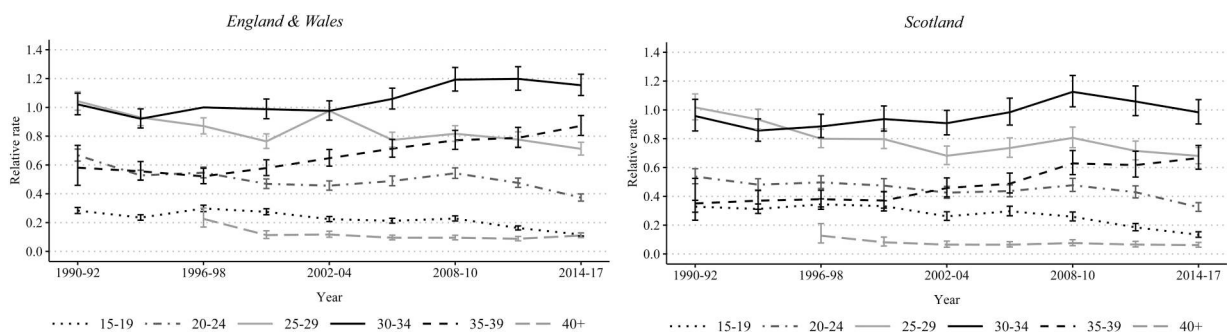


Figure 3 Relative first-birth rates in Britain by women's age, 1990–92 to 2014–17

Notes: Reference group: England & Wales, women aged 30–34 in 1996–98. Vertical bars show 95 per cent CIs. Data for relative first-birth rates for the 40+ group are missing for the 1990–92 period due to small sample sizes.

Source: As for Figure 2.

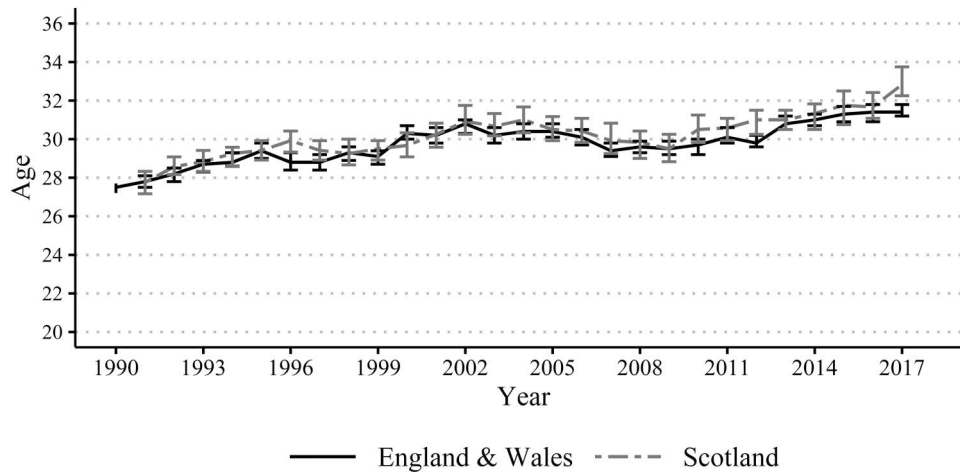


Figure 4 Median age at first birth in Britain by age, 1990–2017

Note: Vertical bars show 95 per cent CIs.

Source: As for Figure 2.

First-birth rates and trends by age are very similar in Scotland.

The trends in average age at first birth present a consistent story: in the early 1990s, median age at first birth was around 28 years for both England & Wales and Scotland (Figure 4). The age at family formation increased in the 1990s. However, this increase stopped in the 2000s, reflecting the stopping of postponement (seen in the same time period in Figure 3). Age at first birth may have even declined slightly in the second half of the 2000s. Since then, median age at first birth has increased further, reaching 31.8 in England & Wales and 32.2 in Scotland in 2017.

Second births

Figure 5 presents relative second-birth rates, standardized for time since first birth and women's age at first birth. Rates are presented relative to fertility in England & Wales in 2000–01. For England & Wales, we see that second-birth rates declined in the 1990s. They increased slightly in the 2000s but have stayed relatively stable since then. The trends are very similar in Scotland; however, the increase in the 2000s was perhaps slightly larger in Scotland than in England & Wales, and it seems that second-birth rates have declined in Scotland in the most recent period. Nevertheless, second-birth rates are

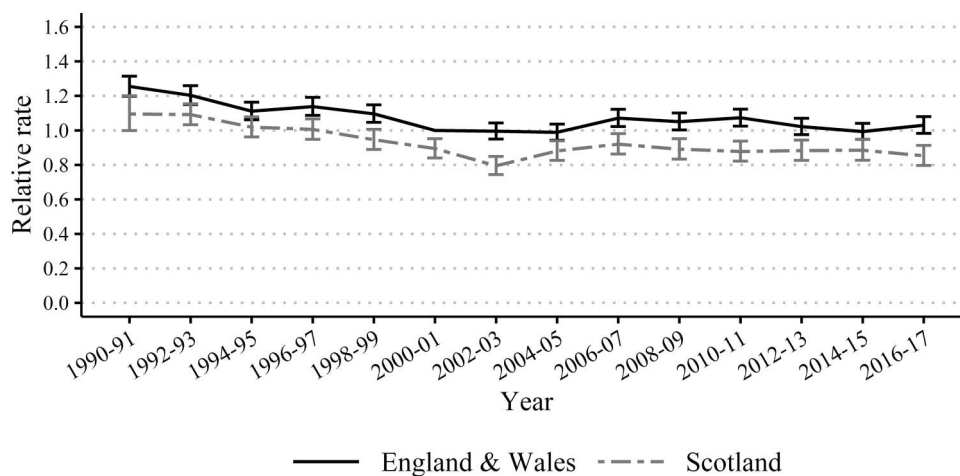


Figure 5 Relative second-birth rates in Britain, 1990–91 to 2016–17 (adjusted for time since first birth and women's age at first birth)

Notes: Reference group: England & Wales, 2000–01. Vertical bars show 95 per cent CIs.

Source: As for Figure 2.

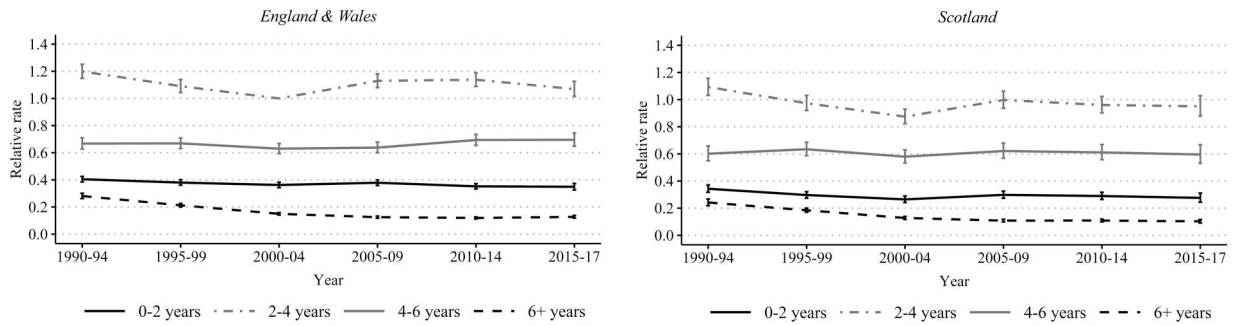


Figure 6 Relative second-birth rates in Britain by time since first birth, 1990–94 to 2015–17 (adjusted for women's age at first birth)

Notes: Reference group: England & Wales, duration 2–4 years, 2000–01. Vertical bars show 95 per cent CIs.

Source: As for Figure 2.

lower in Scotland overall, supporting our expectation that Scottish women are less likely to have a second child than women in England & Wales.

Next, we examine the interval between first and second births, to investigate whether fertility decline occurred because birth intervals changed (i.e. lengthened) in the 1990s or whether all duration-specific rates declined. We see that the most common duration at which women in England & Wales have their second child is between two and four years after their first birth (Figure 6). Interestingly, birth intervals have stayed relatively stable over the years. There is no evidence of lengthening or shortening of intervals (i.e. different time trends by duration since first birth). Clearly, second-birth rates declined at all durations in the 1990s, suggesting that women experienced a declining likelihood of having a second child. Again, the patterns

are relatively similar in Scotland: most common is for women to have their second child between two and four years after their first birth, and birth intervals have stayed relatively stable over the years. However, overall, second-birth rates are lower in Scotland, and the differences between England & Wales and Scotland are more pronounced for second births than for first births.

Third and fourth births

Figure 7 presents trends in third-birth rates, standardized for time since second birth and women's age at second birth. For England & Wales, we see that similarly to first and second births, third-birth rates declined significantly in the 1990s. We observe an increase in third-birth rates after the turn of the

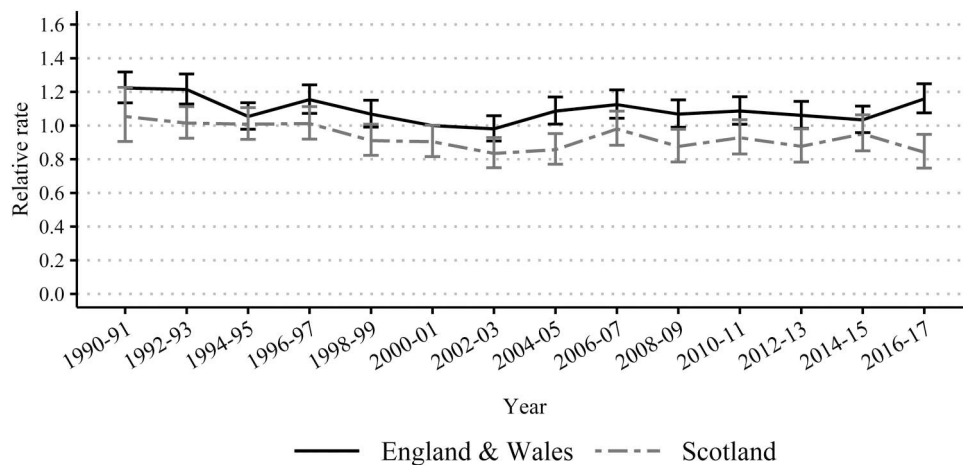


Figure 7 Relative third-birth rates in Britain, 1990–91 to 2016–17 (adjusted for time since second birth and women's age at second birth)

Notes: Reference group: England & Wales, 2000–01. Vertical bars show 95 per cent CIs.

Source: As for Figure 2.

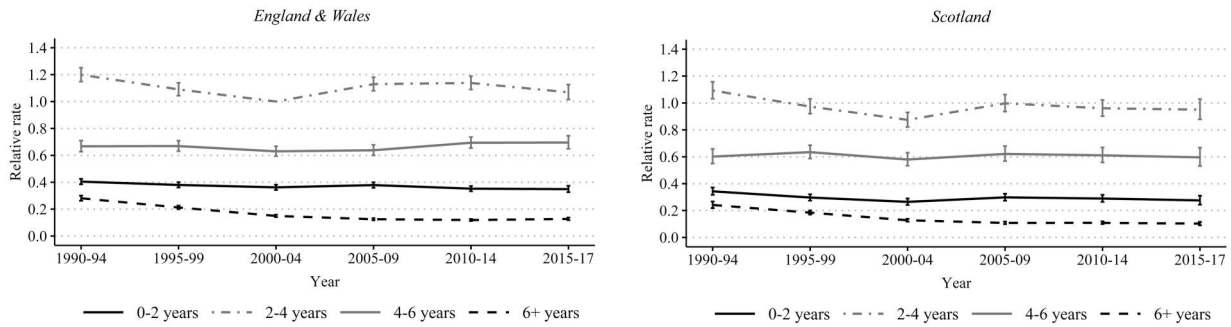


Figure 8 Relative third-birth rates in Britain by time since second birth, 1990–94 to 2015–17 (adjusted for women's age at second birth)

Notes: Reference group: England & Wales, duration 2–4 years, 2000–04. Vertical bars show 95 per cent CIs.

Source: As for Figure 2.

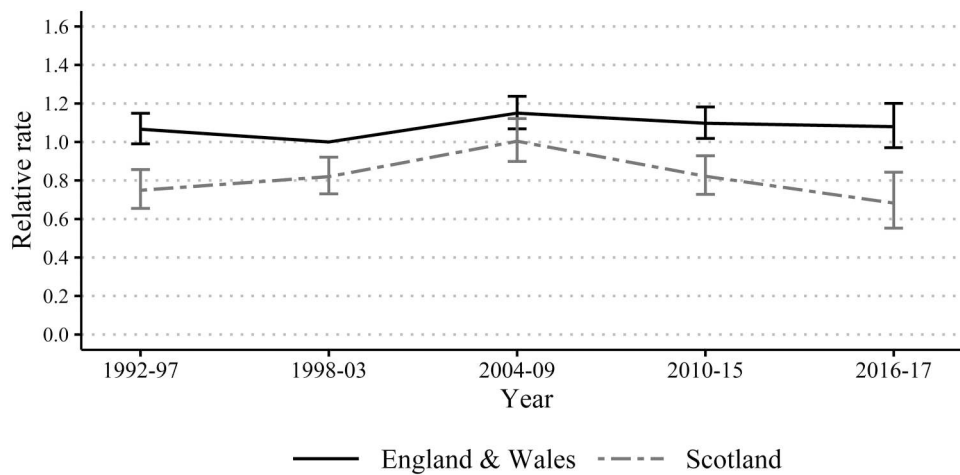


Figure 9 Relative fourth-birth rates in Britain, 1992–97 to 2016–17 (adjusted for time since third birth and women's age at third birth)

Notes: Reference group: England & Wales, 1998–2003. Vertical bars show 95 per cent CIs. Longer time periods are required for fourth-birth rates due to smaller sample sizes compared with other parities.

Source: As for Figure 2.

century and, interestingly, a continued increase in the second half of the 2010s. The trends in third-birth rates are similar in Scotland. The rates declined in the 1990s, increased in the 2000s, and stayed stable in the early 2010s but declined thereafter. Figure 7 also shows that the likelihood of having a third birth is significantly lower in Scotland than in England & Wales.

Do we observe any changes in birth intervals? Again, birth intervals have stayed relatively stable in both England & Wales and Scotland (Figure 8). In summary, third-birth levels declined in the 1990s, showing that women's likelihood of having a third child declined; however, the likelihood increased at the turn of the century. The likelihood of having a third birth is significantly lower in Scotland than in England & Wales (except at longer durations when they are the same).

Finally, Figure 9 presents fourth-birth rates over three decades. The rates are adjusted for time since third birth and women's age at third birth. We see that both England & Wales and Scotland witnessed an increase in fourth-order fertility in the first decade of this century. Since then, there has been a decline, especially for Scotland. Fourth-birth rates are significantly lower in Scotland compared with England & Wales.

Fertility trends in Britain by birth order, ethnic origin, and education

Here, we investigate whether and how much changes in population composition by education and ethnic origin can explain fertility trends over the past

three decades. We also analyse fertility trends within educational and ethnic origin groups.

First births

Figure 10 reports two-yearly first-birth rates with and without adjusting for changes in population composition by education and ethnicity. For England & Wales, we see that the trends over time change slightly with the adjustments, whereas for Scotland they stay almost the same.

Wales the difference between the adjusted and unadjusted estimates shows that the increased share of highly educated women and ethnic minorities (who on average have experienced lower rather than higher first-birth rates in recent years) explain to an extent the lower first-birth rates in the most recent decade. In Scotland, the change across years is negligible, because the share of ethnic minorities is still relatively low.

Figure 11 presents first-birth rates over three decades for three educational groups. We calculate them separately for ages 15–29 and 30–49 because

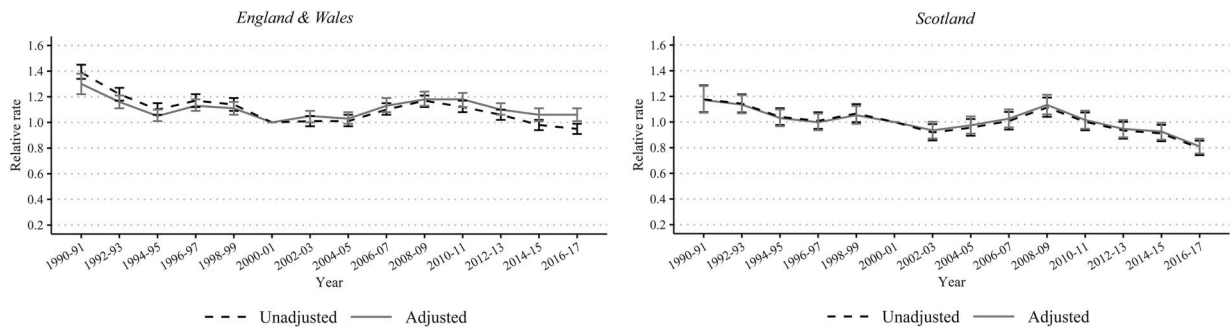
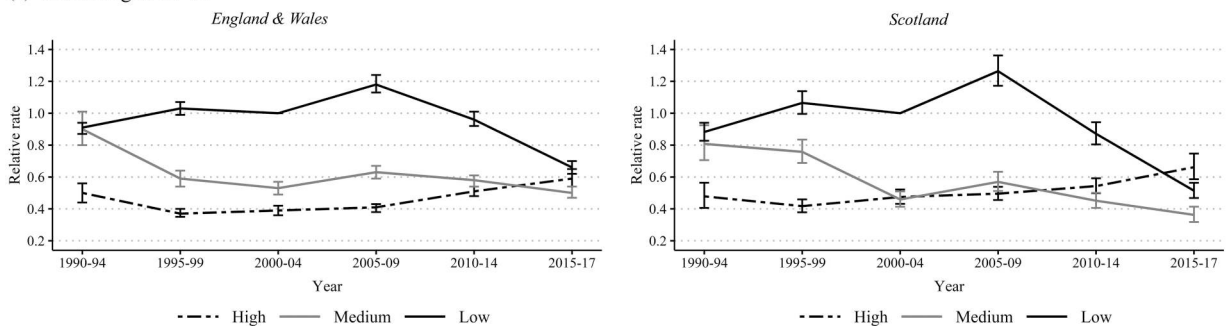


Figure 10 Relative first-birth rates in Britain with and without adjusting for education and ethnicity, 1990–91 to 2016–17 (adjusted for women's age)

Notes: Reference group: 2000–01 for each country. Vertical bars show 95 per cent CIs.

Source: As for Figure 2.

(a) Women aged 15–29



(b) Women aged 30–49

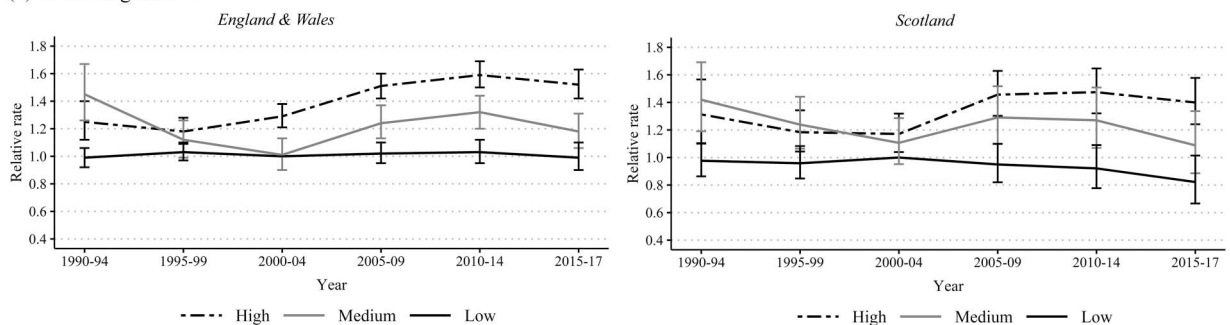


Figure 11 Relative first-birth rates in Britain by education and age group, 1990–94 to 2015–17 (adjusted for women's age)

Notes: Reference group: Low educated, 2000–04 for each country. Vertical bars show 95 per cent CIs.

Source: As for Figure 2.

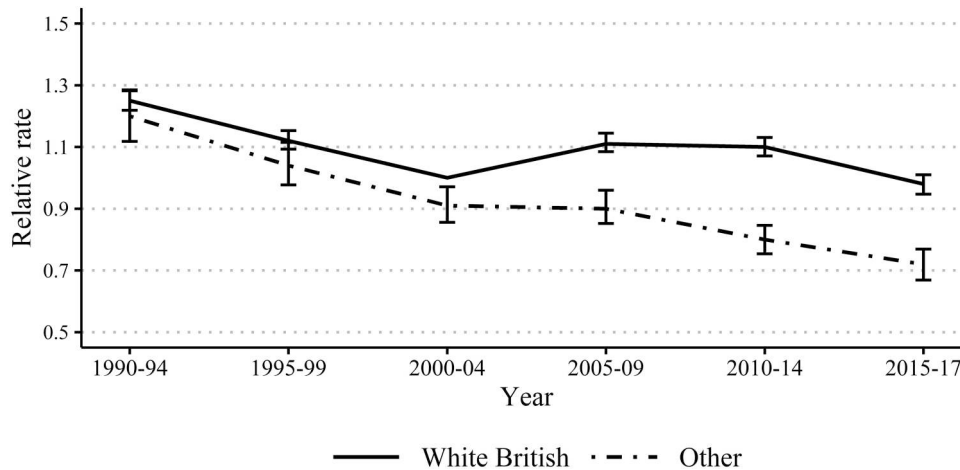


Figure 12 Relative first-birth rates in England & Wales by ethnicity, 1990–94 to 2015–17 (adjusted for women’s age)

Notes: Reference group: White British, 2000–04. Vertical bars show 95 per cent CIs.

Source: Authors’ calculations using ONS LS.

of significant differences in the timing of family formation by educational group that complicate the interpretation of average trends. For both England & Wales and Scotland, we see that low-educated women display the highest and the highly educated the lowest first-birth rates at ages 15–29 (Figure 11(a)). This is largely expected, given that highly educated women start childbearing significantly later than low- and medium-educated women, so we cannot rule out that differences are due solely to timing. We observe two interesting patterns. First, in the early 1990s, first-birth rates among the medium educated were similar to those of low-educated women; their fertility rates declined in the late 1990s and since then have been relatively low (i.e. similar to those of women with high education). However, it is not clear whether this change is real or related to measurement issues (educational level is measured at censuses, i.e. every 10 years). Still, if the former, it is possible that with the expansion of higher education in the second half of the 1990s, many medium-educated women also benefited from the new opportunities and participated in further training. Second, first-birth rates among low-educated women aged 15–29 declined in the 2010s in both England & Wales and Scotland, consistent with theories emphasizing the role of economic uncertainty among those with the least resources.

Patterns of first-birth rates at ages 30–49 by education are largely as expected. For England & Wales, highly educated women exhibit the highest and low-educated women the lowest fertility rates overall (Figure 11(b)). The patterns are similar in Scotland. Regarding trends, the decline in the 1990s was largest among the medium educated,

suggesting that they were responsible for a significant portion of the overall decline in first-birth rates in the 1990s. Interestingly, first-birth rates among the highly educated aged 30–49 increased in the 2000s, suggesting a recuperation among this population subgroup, which may also explain overall fertility increase in that decade. Rates among the low educated remained stable in England & Wales but declined slightly in Scotland in the 2010s. Taking Figure 11(a) and (b) together, we see that first-birth rates among low-educated women in Scotland declined in the most recent decade.

Finally, we investigate trends in first births by ethnic origin by distinguishing: (1) the white British population; and (2) all other ethnicities (including both UK-born individuals and those who immigrated as children). We study only England & Wales, as the ethnic minority population in Scotland is small. Figure 12 shows that first-birth rates were similar among white British and ethnic minority women in the early 1990s. After the beginning of this century, first-birth rates increased significantly among white British women, but there was no such increase among ethnic minority women. As a result, the differences in fertility rates between the two groups increased. Both experienced a fertility decline in the most recent decade, with ethnic minority women showing continuously lower first-birth rates.

Second births

Figure 13 shows two-yearly second-birth rates relative to the 2000–01 period, separately for England

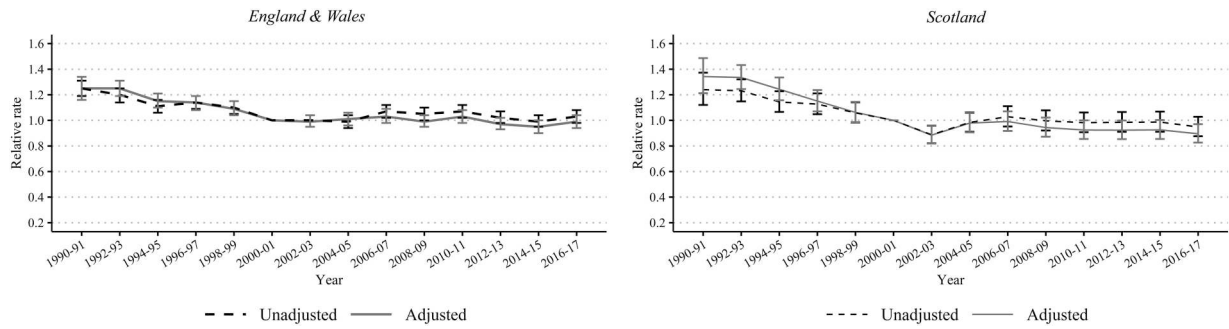


Figure 13 Relative second-birth rates in Britain with and without adjusting for education and ethnicity, 1990–91 to 2016–17 (adjusted for time since first birth and women’s age at first birth)

Notes: Reference group: 2000–01 for each country. Vertical bars show 95 per cent CIs.

Source: As for Figure 2.

& Wales and Scotland. Rates are standardized for time since first birth and women’s age at first birth and are presented with and without adjusting for education and ethnicity. As with first births, we see only marginal changes over time when comparing the results of the two models, emphasizing the limited effect of population composition on second-birth rates. Interestingly, however, the unadjusted rates are slightly lower in the 1990s and higher in the 2010s than the adjusted rates. This is related largely to the changes in population composition by education (i.e. the increased share of population with higher education). However, as we show next, second-birth rates are higher for highly educated women than for the low educated, a pattern which requires further investigation and explanation.

Next, we study second-birth rates by educational level, as shown in Figure 14. For England & Wales, we see that second-birth rates are consistently higher among highly educated women compared with medium- and low-educated women, who

exhibit similar rates to each other. The patterns are relatively similar in Scotland where highly educated women also display the highest second-birth rates. For both cases, we see a decline among medium-educated women in the 1990s, similar to the trends we observed for first births. There may be several reasons why second-birth rates are higher among highly educated women. First, highly educated women tend to have their first child much later than their medium- and low-educated counterparts, and overall, second-birth rates decline with age for everyone. Second, highly educated women may have shorter birth intervals, given that they have children later and have less time to reach desired family size, thus increasing their second-birth rates. Third, highly educated women who become mothers may be a select group and more likely to have a second child than low- or medium-educated mothers, who are less selected into motherhood (Kreyenfeld 2002).

To test the role of age at childbearing, we compared second-birth rates with and without adjusting

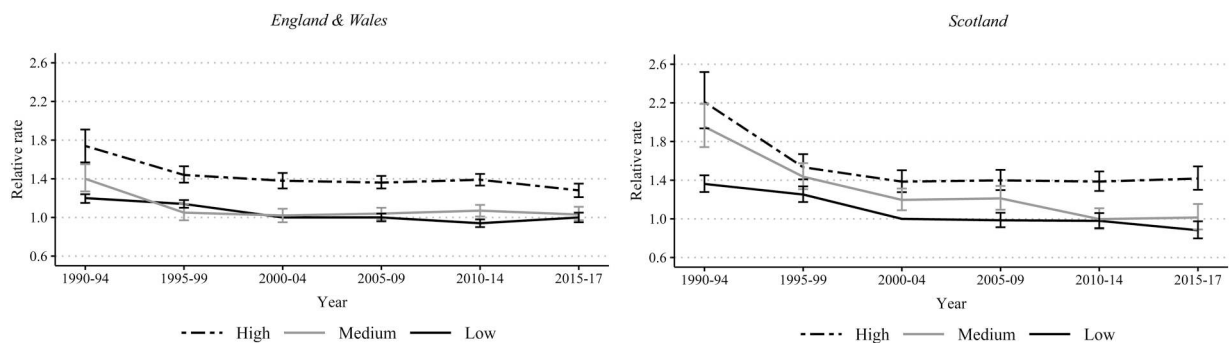


Figure 14 Relative second-birth rates in Britain by education, 1990–94 to 2015–17 (adjusted for time since first birth and women’s age at first birth)

Notes: Reference group: Low educated, 2000–04 for each country. Vertical bars show 95 per cent CIs.

Source: As for Figure 2.

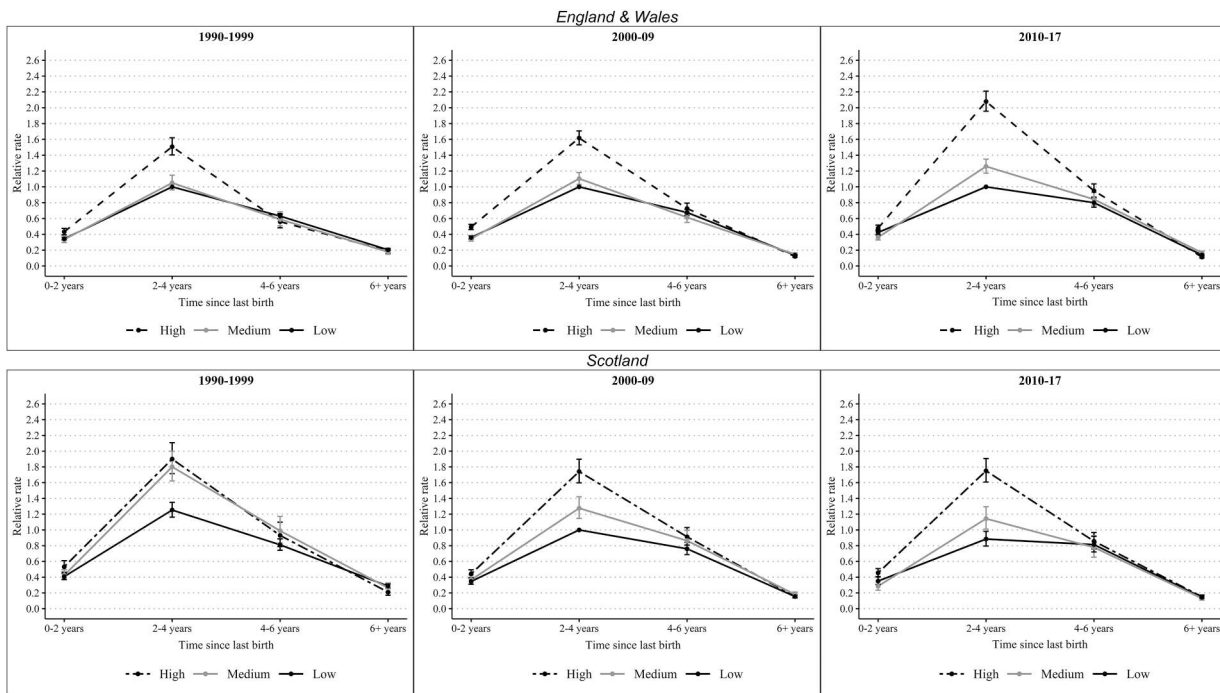


Figure 15 Relative second-birth rates in Britain by education and time since first birth, 1990–99, 2000–09, and 2010–17 (adjusted for women's age at first birth)

Notes: Reference group: Low educated, duration 2–4 years, 2000–09 for each country. Vertical bars show 95 per cent CIs.

Source: As for Figure 2.

for age at first birth (not shown). Once we adjusted the rates for age at first birth, the differences between the groups increased by one-half, suggesting that a significant portion of the higher second-birth rates among the highly educated is attributable to the fact that they have children at (older) ages, when fertility levels are low (e.g. due to lower fecundity). However, further analysis

revealed that their second-birth rates were relatively high even without adjusting for age at first birth. What is the reason: the timing of second births (i.e. shorter birth intervals) or the higher likelihood of having a second child? Figure 15 shows second-birth rates by duration since first birth, education, and time period. For England & Wales, we see that for all three educational groups, second-birth rates

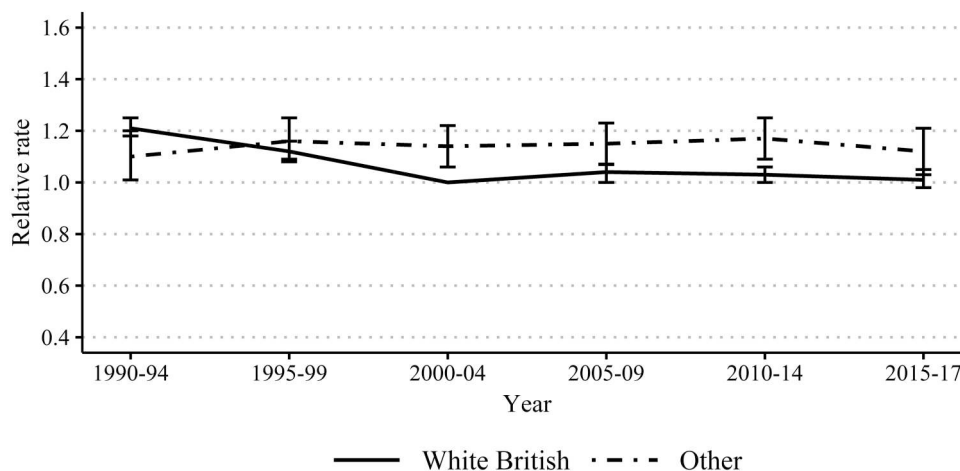


Figure 16 Relative second-birth rates in England & Wales by ethnicity, 1990–94 to 2015–17 (adjusted for time since first birth and women's age at first birth)

Notes: Reference group: White British, 2000–04. Vertical bars show 95 per cent CIs.

Source: As for Figure 12.

follow a rising–falling pattern: they increase and are highest two to four years since first birth and decline thereafter. However, there are some differences across educational groups. Second-birth rates are particularly elevated between two to four years since first birth among highly educated women, suggesting that these women do have shorter birth intervals than medium- and low-educated women. This corresponds to expectations: they have less time to achieve a desired family size (Rendall and Smallwood 2003). At all other durations, their second-birth rates are relatively similar to those of medium- and low-educated women. In brief, second-birth rates for highly educated women are higher because of shorter birth intervals, although there also seems to be a selection effect (i.e. highly educated mothers are more likely to have a second child than medium- and low-educated mothers) (Kravdal 2001).

Finally, we explore trends in second births by ethnic origin in England & Wales. Figure 16 shows that overall, trends are relatively similar among white British and ethnic minority women, except in the early 1990s when second-birth rates were higher (although declining) among white British women. Since the beginning of this century, second-birth rates have been consistently higher among ethnic minority women, compared with the white British population. Further analysis (of birth intervals, not shown) confirmed that the former displayed higher second-birth levels (i.e. the likelihood of mothers to have a second child) than the latter.

Third births

Figure 17 presents third-birth rates in England & Wales and Scotland between 1990–93 and 2015–17. Rates are adjusted for time since second birth and

women's age at second birth. As with first and second births, the patterns change very little once we also adjust the rates for education and ethnicity. This suggests that changes in the educational and ethnic composition of Britain's population have played a negligible (if any) role in fertility dynamics over the last three decades.

Figure 18 shows third-birth rates by educational group, separately for England & Wales and Scotland. For both, we observe higher third-birth rates in the 1990s among highly educated women compared with medium- and low-educated women. However, since the beginning of the twenty-first century, rates have gradually converged across educational groups. It is important to note that these rates are adjusted for age at second birth; if we do not adjust for age at second birth (i.e. include only duration since second birth) then third-birth rates are lower for highly educated women compared with low-educated women.

Again, we investigate possible differences in birth intervals between educational groups and changes over time periods. Figure 19 demonstrates that similar to second births, all educational groups display the highest third-birth rates two to four years since previous birth. However, rates are particularly elevated at this duration for highly educated women, especially in the 1990s, suggesting that the shorter birth intervals largely explain their relatively high third-birth rates in the 1990s and 2000s. Interestingly, there has been a convergence across educational groups in third-birth intervals over time.

Finally, we examine trends in third-birth rates by ethnic group, distinguishing white British and all other groups. Figure 20 shows that third-birth rates were higher for ethnic minority women between 1990 and 2017 compared with white British women. This is very much consistent with previous studies showing that ethnic minority women are

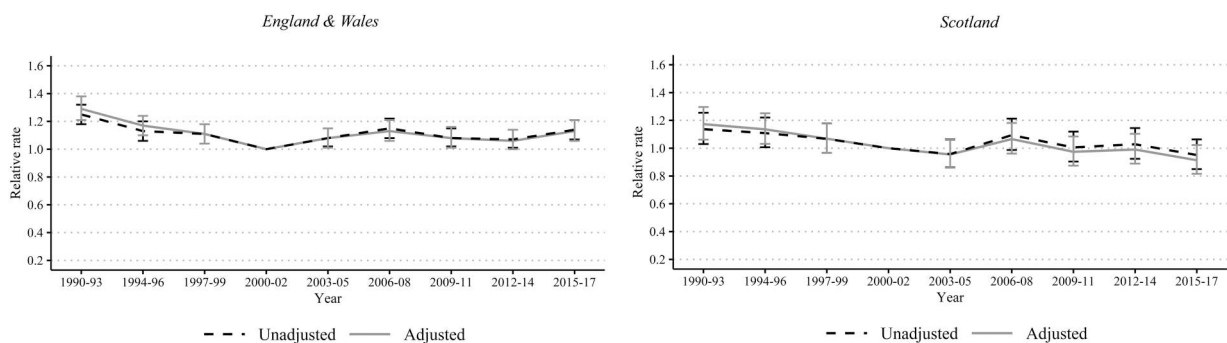


Figure 17 Relative third-birth rates in Britain with and without adjusting for education and ethnicity, 1990–93 to 2015–17 (adjusted for time since second birth and women's age at second birth)

Notes: Reference group: 2000–02 for each country. Vertical bars show 95 per cent CIs.

Source: As for Figure 2.

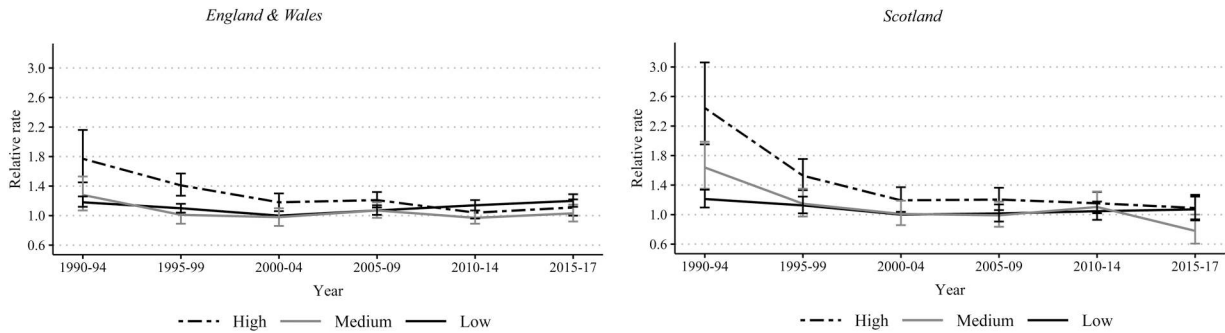


Figure 18 Relative third-birth rates in Britain by education, 1990–94 to 2015–17 (adjusted for time since second birth and women's age at second birth)

Notes: Reference group: Low educated, 2000–04 for each country. Vertical bars show 95 per cent CIs.

Source: As for Figure 2.

more likely to have third births (or larger families) (Kulu et al. 2017). Interestingly, the differences are fairly consistent over the years, and we observe no convergence in third-birth rates between white British and ethnic minority women.

Conclusion

This has been the first study to investigate childbearing in Britain by birth order over three decades and

to compare parity-specific fertility by country. Britain offered us a unique opportunity to compare parity-specific fertility trends across countries that share a similar institutional and socio-economic environment but exhibit distinct national identities. The analysis of the ONS LS and SLS data showed, first, declining first-birth rates in the 1990s, a small increase in the first decade of this century, and a new decline thereafter. Second, the timing of parenthood (postponement or stopping of postponement) rather than changes in childlessness has been

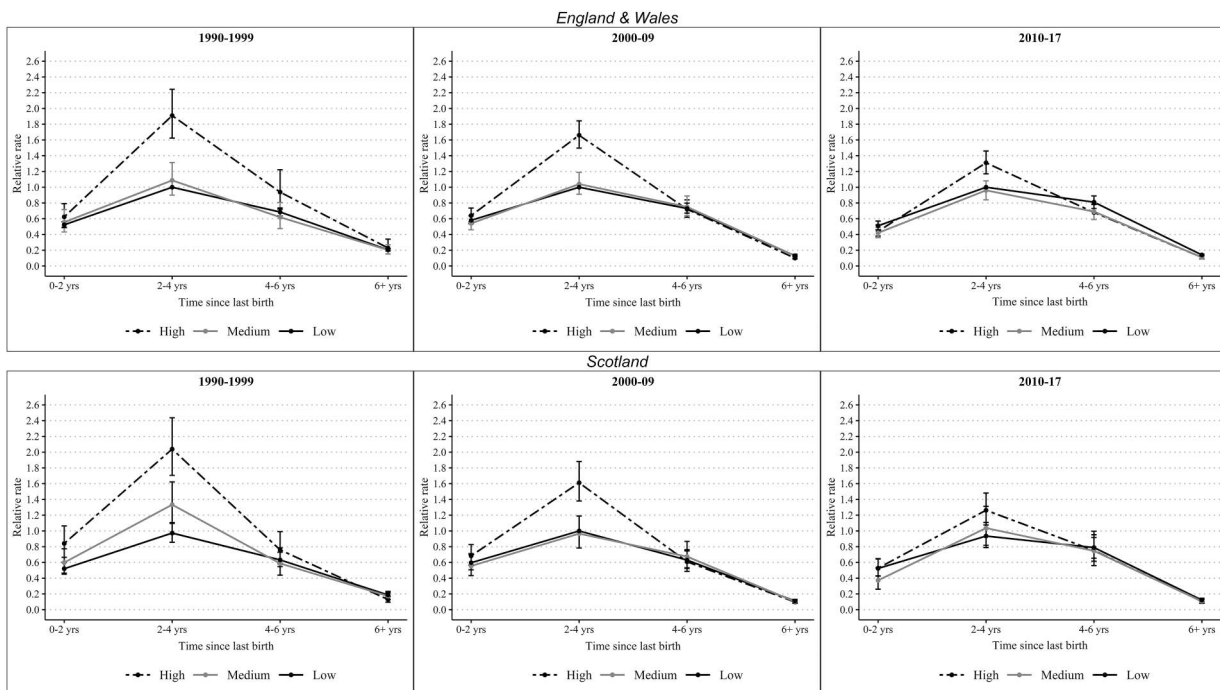


Figure 19 Relative third-birth rates in Britain by education and time since second birth, 1990–99, 2000–09, and 2010–17 (adjusted for women's age at second birth)

Notes: Reference group: Low educated, duration 2–4 years, 2000–09 for each country. Vertical bars show 95 per cent CIs.

Source: As for Figure 2.

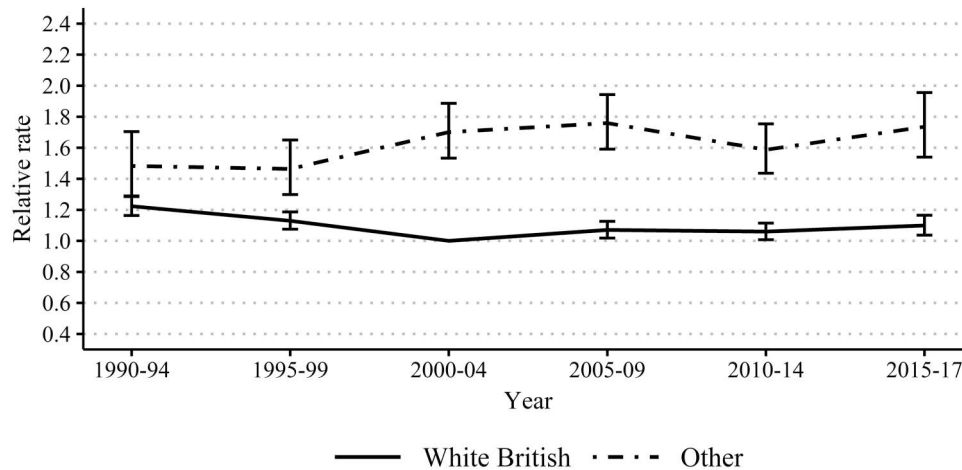


Figure 20 Relative third-birth rates in England & Wales by ethnicity, 1990–94 to 2015–17 (adjusted for time since second birth and women’s age at second birth)

Notes: Reference group: White British, 2000–04. Vertical bars show 95 per cent CIs.

Source: As for Figure 12.

mostly responsible for the changes in first-birth rates. Third, while second- and third-birth rates both declined in the 1990s, second-birth rates remained relatively stable in the first two decades of this century, whereas third-birth rates increased slightly. Fourth, birth intervals have remained relatively stable; therefore, changes in second- and third-birth quantum rather than timing are responsible for the trends in second- and third-birth rates. Fifth, declining first-, second-, and third-birth rates all contributed to declining fertility in the 1990s, whereas changes in first-birth rates explain most of the fluctuations in aggregate fertility since the turn of the century. Sixth, changes in population composition by education and ethnicity explain little of Britain’s recent fertility dynamics. Finally, we observed relatively similar fertility trends and timing in England & Wales and in Scotland but significantly lower second- and third-birth rates in Scotland.

The study largely supports the findings of previous research in Europe that has shown changes in first-birth rates to be the main driver of trends in aggregate period fertility (Hellstrand et al. 2020; Ermisch 2021, 2024; Ohlsson-Wijk and Andersson 2022). Our aim was not to quantify precisely the role of changes in first births in the dynamics of aggregated fertility; nevertheless, it is clear that trends in total fertility have largely followed trends in first-birth rates, especially in the most recent decade. We also expected the timing of family formation to have played the key role in changing first-birth rates. Our analysis suggests that postponement of family formation was responsible for a decline in first-birth rates in the 1990s, whereas the

stopping of this postponement (and even recuperation in the early 2000s) led to an increase during the 2000s. A new wave of postponement since the early 2010s has been behind the current decline in first-birth rates (Ermisch 2021). Importantly, our analysis showed that the recent decline in first-birth rates and concurrent fertility postponement is not surprising: it is part of a long-term trend, which has been ongoing for the last three decades. What is perhaps more surprising is the stopping of postponement in the early 2000s. It is possible that this anomaly was supported by the introduction of family policies and welfare provision in the late 1990s, which helped parents to combine work and childcare (Sigle-Rushton 2008). But this is a topic that requires further research.

The trends in second- and third-birth rates since the turn of the century are also interesting. They show that despite the postponement of family formation, higher-order birth rates have not declined; they have stayed stable or rather increased. At first glance, we might think that when women have their first child at later ages, they may have a second and third child more quickly, inflating second- and third-birth rates. However, our analysis showed that this is not the case: birth intervals have not changed much over the three decades. Second- and third-birth levels have therefore stayed stable or even increased slightly. This suggests that a new pattern has emerged: women are having their first child increasingly later, but once they have a child they are still as likely to have a second and a third child as in the past.

While our findings do not provide definitive evidence as to the underlying explanations of these

changing childbearing patterns, the historical timing of the different waves of fertility postponement is consistent with economic explanations. The increase in age at entry into motherhood in the 1990s occurred in the context of industrial decline and increased job insecurity, best summarized by the notion of ‘risk society’ (Beck 1992). Postponement of first births ceased during a period of economic growth and implementation of policies to help combine work and family (Sigle-Rushton 2008). The second wave of postponement began with the 2008 economic recession, the aftermath of which has continued until recently. However, economic uncertainty would also predict declining second- and third-birth levels, but this was not the case. It is possible that women who have a second and especially a third child are an increasingly select group in low-fertility societies.

The ideational approach would predict increasing postponement of childbearing and smaller family size (Lesthaeghe 2014). We can consider recent fertility trends in Britain as a sign of the emergence of a new behavioural norm, which is much supported by wider structural changes in high-income countries (i.e. longer time spent studying and increased educational levels) and raised awareness of global issues (e.g. climate change). It is possible that this new behavioural norm emerged first among some population groups (e.g. the highly educated) and gradually spread among all (or most) population groups. For example, a decade ago the UK’s age-specific fertility curves were bimodal, but they are not any more, suggesting increasing fertility postponement and declining heterogeneity in the timing of family formation (despite increased socio-cultural heterogeneity in British society) (Tocchioni et al. 2021).

Our analysis of fertility trends by population subgroup has provided further insights into the driving forces behind fertility dynamics. The analysis showed a decline from 2010–14 in first-birth rates among low-educated women at younger ages. These results could be interpreted as postponement for either economic or ideational reasons. The latter approach would predict an increase in first-birth rates among older women in the coming years, whereas under the former approach, the increase would occur only if economic conditions improved. To date, there has been no sign of an increase in first-birth rates at older ages among low-educated women, suggesting that the recent decline in first-birth rates among low-educated women may be related to increased uncertainty among vulnerable groups (i.e. those with weaker

labour market attachment and lower earnings), thus supporting economic theories of fertility change. First-birth rates have been lower for ethnic minorities than for the white British population in the last two decades. At first glance, this seems to lend support to the relevance of economic explanations, especially given that ethnic minorities are normally hit harder during economic crises. However, with a small exception, first-birth rates among ethnic minorities have gradually declined since the early 1990s, suggesting that the decline in family formation (i.e. postponement) is a longer-term trend rather than a recent phenomenon and could equally be driven by long-term economic or ideational changes.

Trends in second- and third-birth rates have been relatively similar across educational and ethnic groups, showing that the group differences between mothers in the likelihood of having a second or third child have persisted over time. We observed some convergence in third-birth rates across educational groups, especially increased similarity in the timing of a third child. In the past, birth intervals were shorter among highly educated women than low- and medium-educated women, but this difference has gradually vanished. Whether this trend is related to government policies (especially increased availability of quality childcare, which has reduced the pressure to have children within a short time interval) or gradually declining family size among the highly educated for ideational or economic reasons remains an open question. Consistently higher third-birth rates among ethnic minorities are consistent with previous research showing that some ethnic minorities have large families: this may be related to their family of origin (i.e. having more siblings), cultural-normative factors (e.g. greater religiosity), or the challenges they face in the labour market (e.g. discrimination) meaning that they focus on childrearing instead (Kulu et al. 2017).

What will happen to fertility rates in the near future? Parity-specific longitudinal data for Britain are currently available only up to 2017, but we know that aggregate fertility has declined further since then, probably because of even further postponement of first births, as the decline has been largest among women in their 20s (Berrington et al. 2022). Continued postponement may eventually lead to an increase in childlessness, as women’s ability to conceive lessens with increasing age. Recent cohort analysis from Finland has shown that some increase in childlessness is likely to occur among cohorts born in the late 1970s and

1980s (Hellstrand et al. 2020). What is going to happen to second and third births in future with the postponement of family formation? As women will have less time to have children, either the likelihood of having a second and third child will decline or women will need to have a second and third child more quickly, suggesting a significant reduction in birth intervals compared with the past. In brief, most literature has focused on the timing and quantum of first births when considering current and future fertility trends, but our study has also emphasized the importance of higher-order births. If first births are increasingly postponed, this will have consequences for second and subsequent births. So far, postponement has had little (if any) effect on the timing or quantum of second and third births, but with the ever increasing age at first birth, this will not necessarily remain the case.

We showed that lower second- and third-birth rates explain the lower total fertility in Scotland compared with England & Wales. Although previous studies have already emphasized the role of higher-order births in fertility differences across Britain's countries, our analysis showed that birth intervals are very similar in Scotland and in England & Wales; hence Scottish women are less likely to have a second or third birth than women in England & Wales. Recent research has shown that there might be some small differences in fertility intentions (i.e. Scottish women hold less certain intentions than those in England & Wales) (Berrington et al. 2023). Our study showed that fertility has been lower in Scotland than in England & Wales because of lower second- and third-birth levels. Interestingly, however, the trends in fertility by birth order have been very similar across Britain's countries over the last three decades, even across population subgroups. This suggests that all countries have been exposed to the same factors driving the long- and short-term fertility trends. It remains an open question whether these factors are so fundamental that their effect outweighs any country differences or whether the countries are sufficiently similar in economy, politics, and culture, despite distinct national identities, that the factors work in a similar way. Both arguments are probably true, given that other high-income countries have experienced a similar rising–falling pattern in period fertility over the last three decades and that there are also differences in the shape and timing of the patterns across countries. The main challenge to policymakers, especially in Scotland, will be to reduce the gap between desired and actual family size for mothers and fathers. With the gradually increasing age at

family formation, this will become an increasingly difficult, although not impossible, task.

Notes and acknowledgements

- 1 Hill Kulu and Sarah Christison are based at the University of St Andrews, Centre for Population Change; Bernice Kuang and Ann Berrington are based at the University of Southampton, Centre for Population Change. Please direct all correspondence to Hill Kulu, University of St Andrews, School of Geography and Sustainable Development, St Andrews KY16 9AL; or by E-mail: hill.kulu@st-andrews.ac.uk.
- 2 Funding: This research was supported by the Economic and Social Research Council (ESRC), grant number ES/S009477/1 (Understanding Recent Fertility Trends in the UK and Improving Methodologies for Fertility Forecasting).
- 3 Acknowledgements and disclaimers: The authors thank Dr Joanne Ellison for her contributions to earlier version of this manuscript. The permission of the Office for National Statistics to use the Longitudinal Study is gratefully acknowledged, as is the help provided by staff at the Centre for Longitudinal Study Information and User Support (CeLSIUS). CeLSIUS is funded by the ESRC under project ES/V003488/1. The authors alone are responsible for the interpretation of the LS data. The help provided by staff at the Longitudinal Studies Centre Scotland (LSCS) is also acknowledged. The LSCS is supported by the ESRC, National Records of Scotland, and the Scottish Government. The authors alone are responsible for the interpretation of the SLS data. Census output is Crown copyright and is reproduced with the permission of the Controller of HMSO and the King's Printer for Scotland.
- 4 Copyright: This work contains statistical data from ONS which is Crown Copyright. The use of the ONS statistical data in this work does not imply the endorsement of ONS in relation to the interpretation or analysis of the statistical data. This work uses research data sets which may not exactly reproduce National Statistics aggregates.
- 5 Ethics: This research was approved by the University of Southampton, Faculty of Social Sciences, Faculty Ethics Committee – ERGO II 56073 on 25 March 2020. It was also approved in principle by the University of St Andrews, School of Geography and Sustainable Development Ethics Committee, acting on behalf of the University Teaching and Research Ethics Committee (due to Covid-19-related travel restrictions), on 24 June 2020.
- 6 All four authors are LS or SLS Approved Researchers; details about the LS and SLS are available on the CALLS-HUB website: <https://calls.ac.uk/guides-resources/>.

Disclosure statement

No potential conflict of interest was reported by the authors.

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